

The Rope Programme

Programme Overview — Assumptions, Structure, Maturity, and Open Problems

The front door to the corpus — generated from the claim registry and computed roadmap

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Programme credit: the core Rope Hypothesis is due to Bill Gaede; this corpus develops and formalises it.

What this programme is

The Rope Hypothesis is a speculative classical model in which interactions are carried by physical strands under tension. This corpus develops its classical continuum sector into a reproducible research programme: a coherent chain from microscopic mechanics up through electromagnetism, optics, and thermodynamics, with every principal claim assigned a status and, wherever possible, backed by an executable benchmark.

The programme’s discipline is its defining feature. Derived results, empirical inputs, and open problems are kept strictly separate; corrections and negative results are preserved as visibly as successes; and the mathematics is kept distinct from the ontology it proposes. This overview is itself generated from the corpus’s machinery, so its figures and tables cannot drift from the papers they summarise.

How to read the programme honestly

The strongest posture here is not any single prediction. It is that the classical core can be independently audited — sector by sector, claim by claim — and that its boundaries are stated rather than blurred. The failed results and open problems below are as much a part of the programme as the derived ones.

Corpus at a glance

Measure	Value
Bundled papers	47
Reproducible benchmarks	21
Registered claims	54 (44 code-backed and machine-verified)
Status distribution	Derived 40, EFT-constrained 1, Modeled 5, Conjecture 3, Failed 3, Open 2

Every code-backed claim can be re-run in one command: `make verify` (runs each benchmark the registry references), or `python tools/verify_corpus.py`.

The classical continuum chain

The programme’s spine is a single geometric structure developed level by level. Each link below is a Mature, benchmark-backed sector:

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Microscopic Mechanics
  -> Homogenization (Gamma-convergence)
    -> Effective Field Theory
      -> Electromagnetism
        -> Classical Optics -> Interface Optics
  -> Defect Theory (2D + 3D)    [same functional]
  -> Statistical Mechanics      [BKT = defect-gas unbinding]

Unified by: Gauge Geometry
(bundle -> connection -> curvature  $F=dA$  -> topology)

```

The gauge-geometry paper is the mathematical reference showing these are one structure; the machine-readable dependency graph (docs/dependency_graph.txt) encodes the links claim by claim.

Sector maturity (computed)

Maturity is computed from each sector's claim statuses and benchmark coverage, not hand-assigned, and cross-checked against stated readiness. There are currently no readiness-vs-evidence flags.

Sector	Maturity	Claims	Derived	Benchmarked
Microscopic Mechanics & Foundations	Mature	13	12	12
Renormalization / EFT	Mature	1	1	1
Electromagnetism	Mature	7	5	7
Classical Optics	Mature	10	10	10
Thermodynamics	Mature	5	4	4
Condensed-Matter Analogues	Developing	1	0	0
Gravity (weak field)	Mature	2	2	2
Solitons & Knot Spectrum	Mature (conditional)	2	1	2
Electroweak (Weinberg angle)	Exploratory	1	0	0
Particle Masses	Exploratory	2	0	0
Gauge Geometry (unification)	Mature	5	4	3
Quantum Boundary (Bell)	Mature (boundary)	5	2	3

Suggested reading order

1. This overview — assumptions, scope, maturity, open problems.
2. Topology and Gauge Geometry Underlying the Rope Programme — the mathematical backbone.
3. Microscopic Mechanics — the endpoint mechanics the chain starts from.
4. A Γ -Convergence Derivation for the Rope Medium — discrete model to continuum functional.

5. Electromagnetism and Classical Optics — the strongest, most self-contained physical sectors.
6. Scope and Limits — what the programme does not claim, especially the quantum boundary.

Open problems and boundaries

The programme marks where it does not yet claim to explain nature. These are surfaced directly from the claim registry, and are treated as first-class content.

Genuinely open — the honest frontier

EM flux tubes as defect lines versus smooth winding (FND-013): a stated open bridge, not claimed.
Amplitude interference from rope dynamics (QB-005): the genuinely open quantum-boundary question.

Documented negative results — kept as findings

Config-counting does not reproduce quantum entanglement (QB-003): the documented quantum boundary.
One-loop fluctuation mass mechanism falsified by $\sim 80\times$ (QB-004).
Lepton ratios with the model's own Weinberg angle fail (PM-002).

Conjectural — suggestive or conditional, not established

The ONTOLOGY itself: that ropes physically instantiate the geometry / F IS the EM field (GG-005) — necessary-but-not-sufficient; the hypothesis under test.
Lepton mass ratios via a Koide phase, only with a measured input (PM-001).
Weinberg angle $\sin^2\theta_W = 1/(3\sqrt{2})$, $\sim 1.94\%$ from measured (EW-001).

How to evaluate this corpus

- Dependency graph — docs/dependency_graph.txt: what rests on what.
- Claim registry — claims.yaml: every claim's status, paper, and benchmark.
- Computed roadmap — docs/roadmap.md: sector maturity, auto-flagged if readiness outruns evidence.
- One-command verification — make verify: runs every code-backed claim's benchmark.
- Heartbeat — make heartbeat: the core validation runs (75/75 + 10/10 + 6/6).

This overview is regenerated by tools/build_overview.py (make overview) from the same registry that drives verification, the dependency graph, and the roadmap — so the front door stays in sync with the corpus behind it.

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